Structured notes: Section 2: charge in C-R and L-C circuits; leakage (Murugeshan 12.3 to 12.5)

Murugeshan, *Electricity and Magnetism* (10th ed.), Ch. 12 §12.1 and §12.2 (Module 4, syllabus item 15–16). Figures are drawn for these notes from the book formulas; the book figure number is in each caption. Problems are not included.

12.3 Growth and decay of charge in a C-R circuit

Source: book-sources/m4-s2-cr-lc.html, Murugeshan pp. 315–317 (printed). Syllabus items 15 and 16. Equations (a)(1)–(a)(3) and (b)(1)–(b)(3) are given in full.

Prerequisites

- $Q = CV$ and the p.d. across a capacitor (pre-physics P3).
- First-order equations solved by separation of variables (pre-math M4).

1. Charging

A capacitor C and a resistance R are connected to an emf E through a key (Fig. 12.6). Q denotes the charge on C at time t, and $I = dQ/dt$.

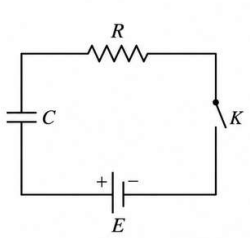


Fig. 12.6 (book): capacitor C, resistance R, Morse key K and cell E in series.

$$E = \frac{Q}{C} + RI \qquad (12.3(a)(1))$$

2. Final charge

At the maximum charge Q_0, $dQ/dt = 0$ and $E = Q_0/C$.

3. Variables separated

$$(Q_0 - Q) = CR\frac{dQ}{dt}$$
$$\frac{dQ}{Q_0 - Q} = \frac{dt}{CR} \qquad (12.3(a)(2))$$

4. Solution

With $Q = 0$ at $t = 0$,

$$Q = Q_0\left(1 - e^{-t/CR}\right) \qquad (12.3(a)(3))$$

The time constant is CR. At $t = CR$ the charge is $0.632\,Q_0$.

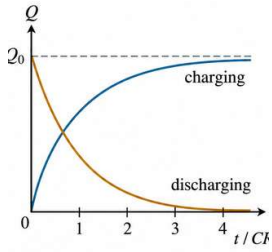


Fig. 12.7 (book): charge against time; growth to Q_0 and discharge from Q_0 are complementary.

5. Discharging

With $E = 0$ and the capacitor charged to Q_0,

$$R\frac{dQ}{dt} + \frac{Q}{C} = 0 \qquad (12.3(b)(1))$$

$$Q = Q_0 e^{-t/CR} \qquad (12.3(b)(2))$$

$$I = \frac{dQ}{dt} = -\frac{Q_0}{CR}e^{-t/CR} = -\frac{Q}{CR} \qquad (12.3(b)(3))$$

At $t = CR$, $Q = 0.368\,Q_0$; the time constant is also the time to fall to $1/e$.

6. Worked examples (printed pp. 316–317)

(1) With $R = 2\ \text{M}\Omega$ and $t = 0.5$ s for $Q/Q_0 = 3/4$, Eq. (a)(3) gives $C = 0.18\ \mu$F. (2) For $C = 0.1\ \mu$F and $R = 10\ \text{M}\Omega$, the p.d. halves in $CR\ln 2 = 0.6931$ s.

12.4 Measurement of high resistance by leakage (background)

Background. *Outside the Module 4 items (15–22). Kept because it applies the discharge law of §12.3. The book numbers no equations in this section. Source: book-sources/m4-s2-cr-lc.html, Murugeshan pp. 318–319 (printed).*

1. Principle

A capacitor of capacitance C and initial charge Q_0 discharges through a resistance R for a time t. From §12.3 the remaining charge is $Q = Q_0 e^{-t/CR}$, so

$$R = \frac{t}{C \log_e (Q_0/Q)} = \frac{t}{2.3026\, C \log_{10} (Q_0/Q)}$$

A high R gives a slow discharge. Measuring Q_0/Q then determines R.

2. Circuit and procedure

C is a capacitor of known value, R the high resistance, B.G. a ballistic galvanometer, E a cell, and K_1, K_2, K_3 tap keys (Fig. 12.8).

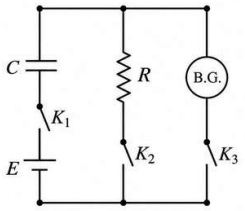


Fig. 12.8 (book): capacitor C, high resistance R, ballistic galvanometer B.G., cell E and keys K1, K2, K3.

(*i*) The capacitor is charged through K_1 and then discharged through the galvanometer, giving a throw $\theta_0 \propto Q_0$. (*ii*) The capacitor is charged again, allowed to leak through R for a time t, and then discharged through the galvanometer, giving a throw $\theta \propto Q$. Since $Q_0/Q = \theta_0/\theta$,

$$R = \frac{t}{2.3026\, C \log_{10} (\theta_0/\theta)}$$

3. Graph

$\log_{10}(\theta_0/\theta)$ plotted against t is a straight line through the origin (Fig. 12.9). Its slope gives $t/\log_{10}(\theta_0/\theta)$, which with C known gives R.

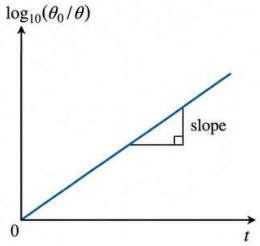


Fig. 12.9 (book): straight line of log10(θ0/θ) against t; the slope gives t / log10(θ0/θ).

4. Worked examples (printed pp. 318–319)

(1) $C = 2\ \mu$F, $R = 100\ \text{M}\Omega$, $Q_0/Q = 2$: $t = 2.3026\, CR \log_{10} 2 = 138.7$ s. (2) $C = 0.2\ \mu$F, throws of 12 cm and 6 cm, $t = 2$ s: $R = 14.43\ \text{M}\Omega$.

12.5 Charging and discharging of a capacitor through an inductance

Source: book-sources/m4-s2-cr-lc.html, Murugeshan pp. 319–321 (printed). Syllabus items 15 and 16. Charging equations are tagged (i)(1)–(4) and discharge equations (ii)(1)–(4), following the book's separate numbering. Section 12.6 onwards is not included.

Prerequisites

- Charge and p.d. of a capacitor; self-inductance and back emf (pre-physics P2, P3).
- Simple harmonic motion (pre-math H2).

1. Charging

Let C and L be joined to an emf E through a key (Fig. 12.10), and Q the charge on C at time t.

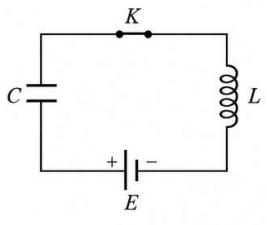


Fig. 12.10 (book): capacitor C and inductance L joined through a Morse key to a cell E.

$$L\frac{dI}{dt} + \frac{Q}{C} = E \qquad (12.5(\text{i})(1))$$

With $I = dQ/dt$, and $Q_0 = CE$, this becomes

$$L\frac{d^2Q}{dt^2} + \frac{Q}{C} = E$$
$$\frac{d^2Q}{dt^2} + \frac{1}{LC}(Q - Q_0) = 0 \qquad (12.5(\text{i})(2))$$

which is simple harmonic motion with period

$$T = 2\pi\sqrt{LC} \qquad (12.5(\text{i})(3))$$

$$\nu = \frac{1}{2\pi\sqrt{LC}} \qquad (12.5(\text{i})(4))$$

2. Discharge

With no external emf (Fig. 12.11),

$$\frac{Q}{C} + L\frac{dI}{dt} = 0$$
$$\frac{d^2Q}{dt^2} + \frac{Q}{LC} = 0 \qquad (12.5(\text{ii})(1))$$

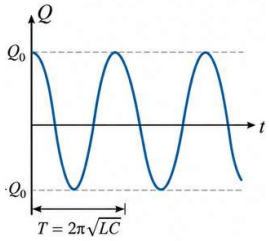
whose general solution, with $\omega = 1/\sqrt{LC}$, is

$$Q = A\sin(\omega t + \theta) \qquad (12.5(\text{ii})(2))$$

With $Q = Q_0$ at $t = 0$ and $\theta = \pi/2$,

$$Q = Q_0 \sin\left(\frac{t}{\sqrt{LC}} + \frac{\pi}{2}\right) \qquad (12.5(\text{ii})(3))$$

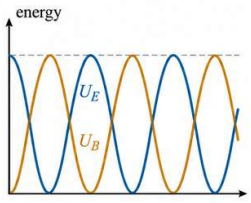
$$\nu = \frac{\omega}{2\pi} = \frac{1}{2\pi\sqrt{LC}} \qquad (12.5(\text{ii})(4))$$



Undamped oscillation of charge, Q = Q_0 sin(t/√LC + π/2), period T = 2π√LC (book, §12.5).

3. Energy

The energy alternates between the electric field of C and the magnetic field of L.



Energy alternates between the electric field of C and the magnetic field of L (book, §12.5).